



CIS-3326-40 Windows Server Admin

Meeting Days, Times, Location:

W 6:00 PM - 7:15 PM

8/19/2024 - 12/10/2024

Armstrong Mabee Business Cntr, 214 Hybrid Course

Course Modality: Hybrid

Semester: Fall, **Year:** 2024

Section(s): CIS-3326-40

Instructor Information

Name: William Nash

Contact Information: (972) 372-4557

Office hours: By Appointment

Course Introduction

Welcome to the Windows Server 2019 course! This comprehensive course will equip you with the knowledge and skills necessary to deploy, configure, and manage Windows Server 2019 environments. You'll explore advanced features, learn about hybrid cloud integration, and gain hands-on experience with essential server administration tasks. Whether you're looking to enhance your IT career or preparing for Microsoft certification, this course will provide you with a solid foundation in Windows Server technologies.

Course Description

This course covers the implementation, management, and maintenance of Windows Server 2019. Topics include server installation, configuration of storage and file systems, networking, virtualization, security, and high availability. The course also explores hybrid scenarios integrating on-premises environments with Azure cloud services. Prerequisites: Basic understanding of networking concepts and familiarity with previous versions of Windows Server.

Course Learning Objectives	Degree Program or GEC Competency Goals
1. Deploy and configure Windows Server 2019 in various scenarios	CIS
2. Implement and manage Active Directory Domain Services	CIS
3. Configure and optimize storage solutions	CIS
4. Implement networking and security features	CIS

Course Learning Objectives	Degree Program or GEC Competency Goals
5. Deploy and manage virtualization using Hyper-V	CIS
6. Implement high availability and disaster recovery solutions	CIS
7. Integrate on-premises environments with Azure cloud services	CIS

Required Materials

Textbook(s)

Title: Windows Server 2019 Inside Out
 Author: Orin Thomas
 Publisher: Microsoft Press
 Year: 2020
 ISBN: 978-0135492277

Software/Apps/Equipment

Windows Server 2019 Evaluation Version (available for free download from Microsoft)
 Virtualization software (e.g., Hyper-V, VirtualBox)
 Azure free account (for hybrid scenarios)

Instruction/Learning Methods and Technologies

The learning methods used in this course include the following:

- x Discussion forums and/or wikis
- x Synchronous (real-time) virtual meetings with peers and/or the instructor
- x Streaming video
- x Quizzes
- x Exams
- x Collaborative assignments
- x Independent homework assignments
- x Writing-intensive assignments
- x Clinical or field work
- x Other: Hands-on lab exercises using virtual machines

Student Workload Expectation

For this 3-credit hour course, students should expect to spend an average of 9 hours per week on course-related activities. This includes 3 hours of instructional time (lectures, demonstrations) and 6 hours of out-of-class work (reading, lab exercises, assignments, exam preparation).

Course Requirements

Submitting and Returning Assignments

Assignment Submission

- Submit all of your assignments via Canvas. [Exceptions include...](#)
- Use the standard Microsoft® Office programs such as Word, PowerPoint, and Excel (available to you through your txwes.edu Office 365 account) to complete your assignments unless assignment directions specify that you should use a different file format.
- For code or configuration files, submit as plain text or within a Word document

Assignment Feedback

- I will provide feedback on assessments within **7** days after submission.
- Feedback will be provided through Canvas comments and rubrics.
- Your grades will be posted to the Canvas Gradebook.

Communication with the Instructor

To contact me with questions and concerns related to this course, please use the Canvas messaging system. I will respond to messages within 24 hours on weekdays and 48 hours on weekends. For urgent matters, you may email me directly at nash@txwes.edu.

Quizzes and Exams

[Quizzes and exams will be administered through Canvas. They will include multiple-choice, true/false, and short answer questions. The final exam will be comprehensive and may include practical exercises.](#)

Evaluation & Grading

All grades will be made available to students in Canvas.

Assignment Category	Number of Graded Items	Point Value per Item	Total Points
Discussions	6	20	120
Assignments	10	50	500
Quizzes	14	20	280
Exams	2	300	300
Total Course Points:			1200

Final Grade	Final Points/Average
A	1080-1200
B	960-1079
C	840-959
D	720-839
F	0-719

Academic Integrity

- Cheating, plagiarism (submitting another person's material as one's own), or completing assignments for another person who will receive academic credit are not permitted. This includes the use of unauthorized books, notebooks, or other sources in order to secure or give help during an examination, the unauthorized copying of examinations, assignments, reports, or term papers, or the presentation of unacknowledged material as if it were the student's own work. Disciplinary action may be taken beyond the academic discipline administered by the course instructor. [Read the University's Academic Integrity Policy.](#)

Class Attendance Policy

Face-to-face Courses

Regular and punctual attendance at all scheduled classes is expected of all students. Students are **required** to notify instructors prior to any missed class and will be held responsible for all class and laboratory assignments.

Online and Hybrid Class

Students enrolled in distance education courses must demonstrate **academic engagement** throughout the course to be counted present.

- **Academic engagement means actively engaging with course content.** Examples of academic engagement include attending a synchronous session, interacting with the instructor, participating in an online discussion, completing interactive tutorials or web-based assignments, and completing assignments or exams.
- Academic engagement **does not include** the act of logging into Canvas, nor does it include participating in academic counseling or advising.

Students must demonstrate academic engagement a minimum of once each week in order to be marked present for that week. During the first week of class, all students will be required to complete either a grade-based or submission-based verification of academic engagement.

University Authorized Absences

- University regulations authorize certain absences of students when representing the University (e.g. participation in intercollegiate athletic competitions, and student government, student development, or fine arts events).
- Absences that may be necessary to fulfill course requirements and are approved by the Provost are also considered to be authorized.
- Students are required to notify instructors prior to any missed class and will be held responsible for all class and laboratory assignments.

Faculty-Initiated Student Withdrawal from a Course

When a student exceeds the permissible number of absences for a class, or, fails to meet the course participation expectations for a class the instructor may drop the student from the class using the Faculty Initiated Student Withdrawal process.

The Registrar will record a grade of "DP" for the course on the student's transcript. The Registrar will record a grade of "WF" (failure) on the student's transcript if the instructor drops the student after the date for receiving a grade of "DP" as specified in the University catalog.

*Note: Per the Office of Financial Aid, students who withdraw from a course **cannot** be re-enrolled in that course during the same semester.*

[The complete Attendance Policy is available in the Catalog.](#)

Other University Policies

- Selected University policies are available in the [Canvas TXWES STUDENT HUB](#). The current Texas Wesleyan University [Catalog](#) and [Student Handbook](#) contain all University policies.

Note: Course syllabi are intended to provide students with basic information concerning the course. The syllabus can be viewed as a "blueprint" for the course. Instructors will inform students of any changes concerning examinations, the grading or attendance policies, or changes in project assignments.

[Syllabi for courses involving laboratories which use chemicals should include the following:](#)

Due to the limitless variety of potentially hazardous chemicals that may be found in a scientific or other laboratory environment, students having known, or potential, health-related concerns, such as allergies, asthma, contact dermatitis, pregnancy, or other physiological sensitivities should check with their physician on the advisability of laboratory work. The instructor and/or chair of the teaching department will provide, upon request, a list of chemicals that will be used in experiments during the semester. More detailed information (that is, Materials Safety Data Sheets) will be furnished by the deans upon written request.

Schedule

Week	Topic	Learning Activities	Assessments
1	Introduction to Windows Server 2019	Read: Ch. 1, Watch: Course Intro Video	Quiz 1, Discussion 1
2	Installing and Configuring Windows Server 2019	Read: Ch. 2, Lab: Server Installation	Quiz 2, Lab 1
3	Implementing Active Directory Domain Services	Read: Ch. 3-4, Lab: AD DS Setup	Quiz 3, Lab 2
4	Implementing Group	Read: Ch. 5, Lab: GPO	Quiz 4, Lab 3,

	Policy	Configuration	Discussion 2
5	Configuring Storage and File Systems	Read: Ch. 6-7, Lab: Storage Spaces	Quiz 5, Lab 4
6	Implementing Networking Services	Read: Ch. 8-9, Lab: DNS and DHCP	Quiz 6, Lab 5
7	Implementing Remote Access	Read: Ch. 10, Lab: VPN Setup	Quiz 7, Lab 6, Discussion 3
8	Midterm Project	Review Weeks 1-7	Midterm Project Due
9	Implementing Hyper-V	Read: Ch. 11-12, Lab: VM Deployment	Quiz 8, Lab 7
10	Implementing Containers	Read: Ch. 13, Lab: Docker on Windows	Quiz 9, Lab 8, Discussion 4
11	Implementing High Availability	Read: Ch. 14, Lab: Failover Clustering	Quiz 10, Lab 9
12	Implementing Disaster Recovery	Read: Ch. 15, Lab: Backup and Restore	Quiz 11, Lab 10
13	Implementing Update Management	Read: Ch. 16, Lab: WSUS Configuration	Quiz 12, Discussion 5
14	Monitoring and Troubleshooting	Read: Ch. 17, Lab: Performance Monitoring	Quiz 13
15	Integrating with Azure	Read: Ch. 18, Lab: Azure AD Connect	Quiz 14, Discussion 6
16	Final Exam	Final exam	Final Exam

Additional Course Information

Lab Environment

This course requires hands-on experience with Windows Server 2019. Students will need access to a lab environment, which can be set up using one of the following methods:

1. On-campus computer lab (if available)
2. Personal computer with virtualization software (e.g., Hyper-V, VirtualBox)
3. Cloud-based lab environment (e.g., Azure labs, AWS Educate)

Detailed instructions for setting up your lab environment will be provided in the first week of the course.

Microsoft Certification

This course aligns with the Microsoft Certified: Windows Server Hybrid Administrator Associate certification. While obtaining the certification is not required for this course, students are encouraged to consider taking the AZ-800 (Administering Windows Server Hybrid Core Infrastructure) and AZ-801 (Configuring Windows Server Hybrid Advanced Services) exams after completing this course.

Detailed Assignment Descriptions

Weekly Quizzes (20 points each, 280 points total)

- Short, timed quizzes covering the week's material
- Multiple choice and true/false questions
- Designed to reinforce key concepts and prepare for the final exam

Lab Assignments (50 points each, 500 points total)

- Hands-on exercises performed in your lab environment
- Step-by-step instructions provided
- Submit screenshots and/or configuration files as proof of completion
- Graded on accuracy and completeness

Midterm Project (100 points)

- Design and implement a small business network using Windows Server 2019
- Include Active Directory, file services, and basic networking
- Submit a written report and demonstration video

Final Exam (200 points)

- Comprehensive exam covering all course material
- Mix of multiple choice, short answer, and practical exercises
- Timed exam to be completed within a specified window

Discussions (20 points each, 120 points total)

- Participate in online discussions about course topics
- Initial post due by Wednesday, two response posts due by Sunday each week
- Graded on quality of contribution, critical thinking, and engagement with peers

Late Work Policy

Assignments submitted after the due date will be subject to a 10% penalty per day late, up to a maximum of 3 days. After 3 days, assignments will not be accepted without prior arrangement with the instructor. Extensions may be granted in extenuating circumstances if requested in advance.

Technical Support

For technical issues related to Canvas, please contact the IT Help Desk at [phone number] or [email]. For issues with the lab environment or course software, please post in the dedicated "Technical Support" forum in Canvas, where the instructor and fellow students can assist you.

Course Materials Copyright Statement

The materials used in connection with this course may be subject to copyright protection and are only for the use of students officially enrolled in the course for the educational purposes associated with the course. Copyright law must be considered before copying, retaining, or disseminating materials outside of the course.

Accessibility Statement

Texas Wesleyan University is committed to providing equal access to all students. If you have a documented disability and require accommodations to obtain equal access in this course, please contact the instructor at the beginning of the semester or when given an assignment for which an accommodation is required. Students with disabilities must verify their eligibility through the Office of Disability Services.

Diversity and Inclusion Statement

This course is designed to support an inclusive learning environment where diverse perspectives are recognized, respected, and seen as a source of strength. It is our intent to present materials and activities that are respectful of diversity: gender identity, sexuality, disability, age, socioeconomic status, ethnicity, race, nationality, religion, and culture. Your suggestions are encouraged and appreciated.